Practice 2–3 — Relations

Solutions to Selected Problems

Problems 1.2, 2.1, 2.2, 3.3, 4.1, 7.2

Discrete Mathematics — QYSU

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1 Problem 1.2 — Domain and Range of Relations

Theory Recap: Domain and Range

For a relation $\alpha \subseteq A \times B$:

- **Domain:** $D_\alpha = \{\, x \mid \exists\, y :\ (x, y) \in \alpha \,\}$ — the set of all first coordinates.
- **Range:** $R_\alpha = \{\, y \mid \exists\, x :\ (x, y) \in \alpha \,\}$ — the set of all second coordinates.

Problem Statement

Find the domain D_α and range R_α for each of the following relations.

(a) $\alpha = \{(a, 1),\ (a, 2),\ (c, 1),\ (c, 2),\ (c, 4),\ (d, 5)\}$

Reading off the first coordinates and second coordinates:

$$D_\alpha = \{a,\ c,\ d\}, \qquad R_\alpha = \{1,\ 2,\ 4,\ 5\}.$$

(b) $\alpha = \{(1, 2),\ (2, 4),\ (3, 6),\ (4, 8),\ \dots\}$

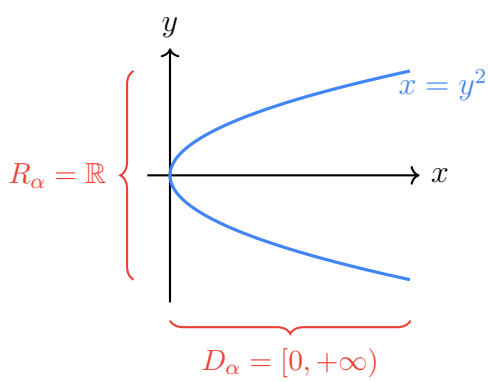
The pattern is $(n,\ 2n)$ for $n \in \mathbb{N}$, so:

$$D_\alpha = \mathbb{N} = \{1, 2, 3, \dots\}, \qquad R_\alpha = \{2, 4, 6, 8, \dots\} = 2\mathbb{N}.$$

(c) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x = y^2\}$

Since $x = y^2 \geq 0$ for all y, and for every $x \geq 0$ there exists $y = \pm\sqrt{x}$:

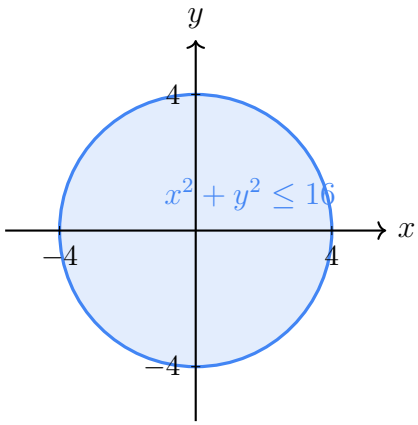
$$D_\alpha = [0, +\infty), \qquad R_\alpha = \mathbb{R}.$$



(d) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 16\}$

This is a closed disk of radius 4 centred at the origin. For any x with $|x| \leq 4$, there exists y satisfying $x^2 + y^2 \leq 16$, and vice versa.

$D_\alpha = [-4,\, 4], \qquad R_\alpha = [-4,\, 4].$



(e) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid x \mid y\}$

Here $x \mid y$ means "x divides y." Every $x \in \mathbb{N}$ divides some y (e.g. $y = x$), so $D_\alpha = \mathbb{N}$. Every $y \in \mathbb{N}$ is divided by some x (e.g. $x = 1$), so $R_\alpha = \mathbb{N}$.

$D_\alpha = \mathbb{N}, \quad R_\alpha = \mathbb{N}.$

(f) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid y \mid x\}$

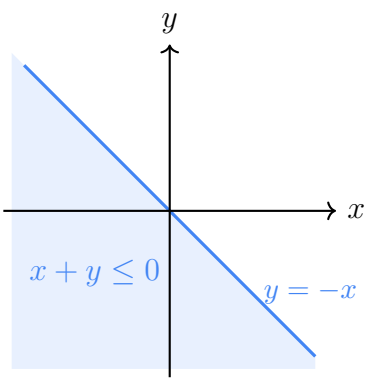
This is the inverse of part (e). By the same reasoning:

$D_\alpha = \mathbb{N}, \quad R_\alpha = \mathbb{N}.$

(g) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 0\}$

This is the half-plane below and including the line $y = -x$. For any $x \in \mathbb{R}$, choose $y = -x - 1$; then $x + y = -1 \leq 0$. For any $y \in \mathbb{R}$, choose $x = -y - 1$.

$D_\alpha = \mathbb{R}, \quad R_\alpha = \mathbb{R}.$



(h) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid 2x \geq 3y\}$

Equivalently $y \leq \frac{2}{3}x$. For any x, choose $y = 0$ (when $x \geq 0$) or any sufficiently small y. For any y, choose $x = \frac{3}{2}y$.

$D_\alpha = \mathbb{R}, \quad R_\alpha = \mathbb{R}.$

2 Problem 2.1 — Graph and Matrix of Relations

Theory Recap: Graphs and Matrices of Relations

Let α be a relation on a finite set $A = \{a_1, \ldots, a_n\}$.

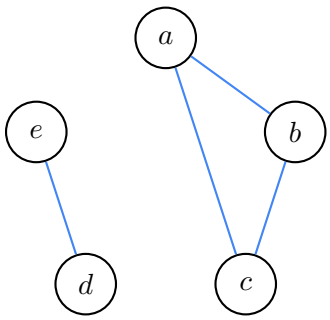
- **Graph:** Each element of A is a vertex. For a symmetric relation, if $(a_i, a_j) \in \alpha$ with $i \neq j$, draw an *undirected* edge between a_i and a_j.
- **Matrix:** The $n \times n$ matrix M with $M_{ij} = 1$ if $(a_i, a_j) \in \alpha$, and $M_{ij} = 0$ otherwise. For a symmetric relation $M = M^T$.

Problem Statement

Construct the graph and matrix for the following relations on $A = \{a, b, c, d, e\}$.

(a) $\alpha = \{(a,b), (b,a), (b,c), (c,b), (c,a), (a,c), (d,e), (e,d)\}$

Since every pair appears with its reverse, the relation is symmetric. Undirected edges: a–b, b–c, a–c, d–e.

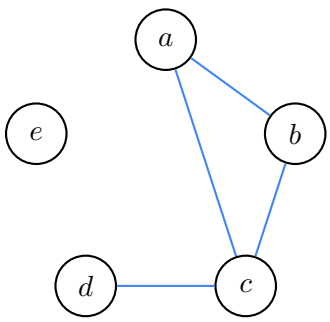


$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

Two connected components: the triangle $\{a, b, c\}$ and the edge $\{d, e\}$.

(b) $\alpha = \{(a,b), (b,a), (b,c), (c,b), (c,d), (d,c), (c,a), (a,c)\}$

Undirected edges: a–b, b–c, c–d, a–c.



$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

Triangle $\{a, b, c\}$ with d attached to c; vertex e is isolated.

(c)

Part (c) is identical to part (b) in the original problem sheet. The graph and matrix are the same as above.

3 Problem 2.2 — Digraph and Matrix of Relations

Theory Recap: Directed Graphs (Digraphs)

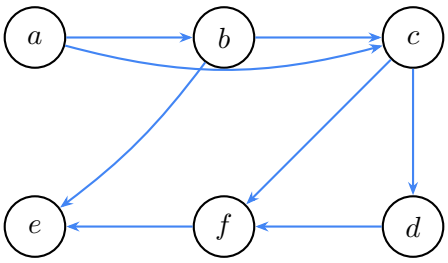
For a (not necessarily symmetric) relation α on A:

- **Digraph:** Vertices are elements of A. For each $(a_i, a_j) \in \alpha$, draw an arrow $a_i \to a_j$. A pair (a_i, a_i) is a *loop*.
- **Matrix:** Same construction; the matrix need not be symmetric.

Problem Statement

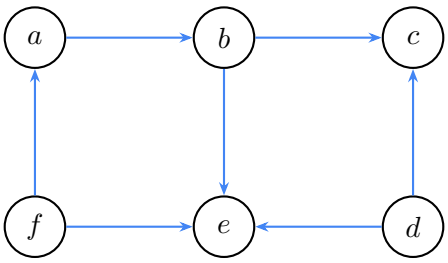
Construct the digraph and matrix for the following relations on $A = \{a, b, c, d, e, f\}$.

(a) $\alpha = \{(a,b), (b,c), (a,c), (b,e), (c,f), (c,d), (d,f), (f,e)\}$



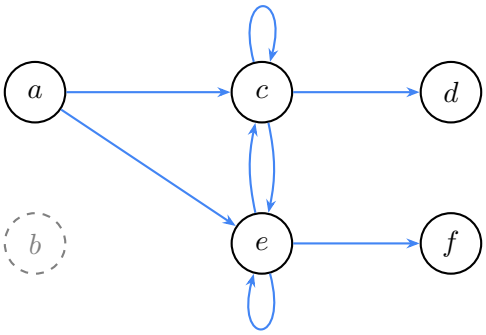
$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

(b) $\alpha = \{(a,b), (b,c), (d,c), (d,e), (f,e), (f,a), (b,e)\}$



$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

(c) $\alpha = \{(a,c),(c,c),(c,d),(c,e),(e,c),(e,e),(e,f),(a,e)\}$



Note: vertex b (dashed) is completely isolated — no edges touch it.

$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

The loops at c and e appear as 1's on the main diagonal. The b-row and b-column are all zeros.

4 Problem 3.3 — Composition of Relations

Theory Recap: Composition of Relations

Let $\alpha \subseteq A \times B$ and $\beta \subseteq B \times C$. The **composition** $\alpha \circ \beta$ (also written $\alpha \cdot \beta$) is the relation on $A \times C$ defined by:

$$(x, z) \in \alpha \circ \beta \iff \exists\, y \in B : \ (x, y) \in \alpha \ \wedge \ (y, z) \in \beta.$$

Read left-to-right: first go through α (from x to some y), then through β (from y to z).

Problem Statement

Let α and β be relations on $\mathbb{Z}$:

$$\alpha = \{(x,\, x+2) \mid x \in \mathbb{Z}\}, \qquad \beta = \{(x,\, x^2) \mid x \in \mathbb{Z}\}.$$

Find $\alpha \circ \beta$ and $\beta \circ \alpha$.

Finding $\alpha \circ \beta$

We apply α first, then β:

- $(x, y) \in \alpha \implies y = x + 2$.
- $(y, z) \in \beta \implies z = y^2 = (x+2)^2$.

$$\alpha \circ \beta = \{\, (x,\ (x+2)^2) \mid x \in \mathbb{Z} \,\} = \{\, (x,\ x^2 + 4x + 4) \mid x \in \mathbb{Z} \,\}.$$

Finding $\beta \circ \alpha$

We apply β first, then α:

- $(x, y) \in \beta \implies y = x^2$.
- $(y, z) \in \alpha \implies z = y + 2 = x^2 + 2$.

$$\beta \circ \alpha = \{\, (x,\ x^2 + 2) \mid x \in \mathbb{Z} \,\}.$$

Verification and comparison

x	$\alpha \circ \beta$: $(x+2)^2$	$\beta \circ \alpha$: $x^2 + 2$
-2	0	6
0	4	2
1	9	3
3	25	11

Key observation: $\alpha \circ \beta \neq \beta \circ \alpha$, confirming that composition of relations is *not commutative* in general.

$$\begin{tikzcd} x \arrow[r, "\alpha"] & x+2 \arrow[r, "\beta"] & (x+2)^2 \end{tikzcd}$$

$$\alpha \circ \beta$$

$$\begin{tikzcd} x \arrow[r, "\beta"] & x^2 \arrow[r, "\alpha"] & x^2+2 \end{tikzcd}$$

$$\beta \circ \alpha$$

5 Problem 4.1 — Properties of Relations

Theory Recap: Properties of Relations

A relation α on a set A may satisfy:

- **Reflexive:** $\forall x \in A: \ (x, x) \in \alpha$.
- **Symmetric:** $(x, y) \in \alpha \ \Rightarrow \ (y, x) \in \alpha$.
- **Antisymmetric:** $(x, y) \in \alpha \ \wedge \ (y, x) \in \alpha \ \Rightarrow \ x = y$.
- **Transitive:** $(x, y) \in \alpha \ \wedge \ (y, z) \in \alpha \ \Rightarrow \ (x, z) \in \alpha$.

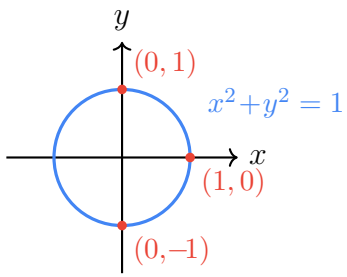
Note that symmetric and antisymmetric are *not* negations of each other — a relation can be both (e.g. Δ_A) or neither.

Problem Statement

Determine which properties each of the following relations possesses.

(a) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$

Geometrically, $(x, y) \in \alpha$ iff the point (x, y) lies on the unit circle.



Reflexive? No. We'd need $2x^2 = 1$ for *all* x, which fails for $x = 0$.

Symmetric? Yes. $x^2 + y^2 = y^2 + x^2$. ✓

Antisymmetric? No. $(0, 1) \in \alpha$ and $(1, 0) \in \alpha$, but $0 \neq 1$.

Transitive? No. $(1, 0)$ and $(0, 1) \in \alpha$, but $(1, 1) \notin \alpha$ since $1 + 1 = 2 \neq 1$.

Symmetric only.

(b) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y^2\}$

Equivalently $|x| = |y|$, so $y = \pm x$.

Reflexive? Yes. $x^2 = x^2$ always holds. ✓

Symmetric? Yes. $x^2 = y^2 \Rightarrow y^2 = x^2$. ✓

Antisymmetric? No. $1^2 = (-1)^2$ and $(-1)^2 = 1^2$, but $1 \neq -1$.

Transitive? Yes. $x^2 = y^2$ and $y^2 = z^2$ imply $x^2 = z^2$. ✓

Reflexive, symmetric, transitive — this is an **equivalence relation**.
Equivalence classes: $[x] = \{x, -x\}$ for each $x \in \mathbb{R}$.

(c) $\alpha = \{(x, y) \in \mathbb{Z}^2 \mid x \mid y\}$

Here $x \mid y$ means "x divides y," i.e. $\exists\, k \in \mathbb{Z} :\ y = kx$.

Reflexive? Yes. $x = 1 \cdot x$, so $x \mid x$ for all $x \in \mathbb{Z}$. ✓

Symmetric? No. $1 \mid 2$ but $2 \nmid 1$.

Antisymmetric? No (on $\mathbb{Z}$!). We have $2 \mid (-2)$ and $(-2) \mid 2$, but $2 \neq -2$.

Transitive? Yes. If $y = kx$ and $z = \ell y$, then $z = \ell k x$, so $x \mid z$. ✓

Reflexive and transitive, but *not* antisymmetric on $\mathbb{Z}$ (due to sign).

(d) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid x \mid y\}$

Same as (c) but restricted to $\mathbb{N} = \{1, 2, 3, \dots\}$ (positive integers).

Reflexive? Yes. $x = 1 \cdot x$. ✓

Symmetric? No. $1 \mid 2$ but $2 \nmid 1$.

Antisymmetric? Yes. If $x \mid y$ and $y \mid x$ with $x, y > 0$: then $y = kx$ and $x = \ell y$ with $k, \ell \geq 1$, so $x = k\ell\, x$ giving $k\ell = 1$, hence $k = \ell = 1$ and $x = y$. ✓

Transitive? Yes (same argument as (c)). ✓

Reflexive, antisymmetric, transitive — this is a **partial order**.
Compare with (c): restricting from $\mathbb{Z}$ to $\mathbb{N}$ restores antisymmetry.

(e) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid \gcd(x, y) = 1\}$

$(x, y) \in \alpha$ iff x and y are coprime.

Reflexive? No. $\gcd(2, 2) = 2 \neq 1$.

Symmetric? Yes. $\gcd(x, y) = \gcd(y, x)$. ✓

Antisymmetric? No. $\gcd(2, 3) = 1$ and $\gcd(3, 2) = 1$, but $2 \neq 3$.

Transitive? No. $\gcd(2, 3) = 1$ and $\gcd(3, 4) = 1$, but $\gcd(2, 4) = 2 \neq 1$.

Symmetric only.

(f) $\alpha = \{(x, y) \in \mathbb{Z}^2 \mid \gcd(x, y) = 1\}$

The analysis is identical to part (e) since gcd depends only on absolute values. The same counterexamples apply.

Symmetric only.

Summary Table

Relation	Refl.	Symm.	Antisymm.	Trans.	Type
(a) $x^2 + y^2 = 1$ on $\mathbb{R}$	—	✓	—	—	—
(b) $x^2 = y^2$ on $\mathbb{R}$	✓	✓	—	✓	equivalence
(c) $x \mid y$ on $\mathbb{Z}$	✓	—	—	✓	preorder
(d) $x \mid y$ on $\mathbb{N}$	✓	—	✓	✓	partial order
(e) $\gcd = 1$ on $\mathbb{N}$	—	✓	—	—	—
(f) $\gcd = 1$ on $\mathbb{Z}$	—	✓	—	—	—

6 Problem 7.2 — Closures of a Relation

Theory Recap: Closures of a Relation

Let α be a relation on a finite set A.

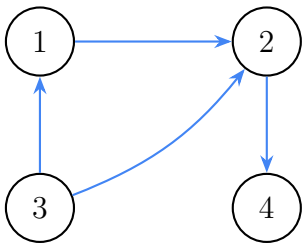
- **Reflexive closure** $r(\alpha) = \alpha \cup \Delta_A$, where $\Delta_A = \{(x, x) \mid x \in A\}$.
- **Symmetric closure** $s(\alpha) = \alpha \cup \alpha^{-1}$, where $\alpha^{-1} = \{(y, x) \mid (x, y) \in \alpha\}$.
- **Transitive closure** $t(\alpha) = \alpha \cup \alpha^2 \cup \alpha^3 \cup \cdots$ (union of all powers until no new pairs appear).

The closure adds the *fewest* pairs needed to acquire the property.

Problem Statement

Let $A = \{1, 2, 3, 4\}$ and $\alpha = \{(1, 2), (3, 1), (3, 2), (2, 4)\}$.
Find $r(\alpha)$, $s(\alpha)$, and $t(\alpha)$.

The original relation



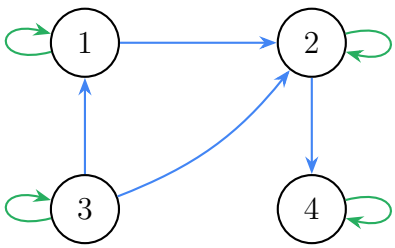
$$M_\alpha = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Rows/columns: $1, 2, 3, 4$.

Reflexive closure $r(\alpha)$

Add all diagonal pairs (x, x) for $x \in A$:

$$\begin{aligned} r(\alpha) &= \alpha \cup \{(1, 1), (2, 2), (3, 3), (4, 4)\} \\ &= \{(1, 1), (1, 2), (2, 2), (2, 4), (3, 1), (3, 2), (3, 3), (4, 4)\}. \end{aligned}$$



$$M_{r(\alpha)} = \begin{pmatrix} \mathbf{1} & 1 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 1 \\ 1 & 1 & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{pmatrix}$$

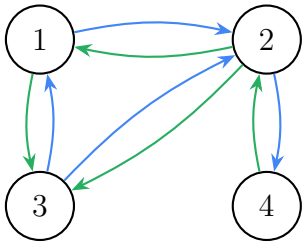
Bold = added 1's on diagonal.

Symmetric closure $s(\alpha)$

Add the reverse of every pair:

$$\alpha^{-1} = \{(2, 1), (1, 3), (2, 3), (4, 2)\}.$$

$$s(\alpha) = \alpha \cup \alpha^{-1} = \{(1, 2), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 2)\}.$$



$$M_{s(\alpha)} = \begin{pmatrix} 0 & 1 & \mathbf{1} & 0 \\ \mathbf{1} & 0 & \mathbf{1} & 1 \\ 1 & 1 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \end{pmatrix}$$

Bold = added 1's (M is now symmetric).

Transitive closure $t(\alpha)$

We compute successive powers of α:

Step 1: $\alpha^2 = \alpha \circ \alpha$**.** For each chain $x \xrightarrow{\alpha} y \xrightarrow{\alpha} z$:

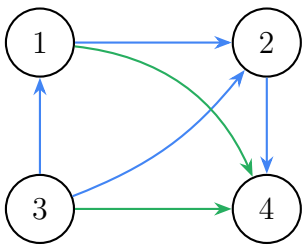
- $(1, 2)$ and $(2, 4)$: gives $\mathbf{(1, 4)}$. **New!**
- $(3, 1)$ and $(1, 2)$: gives $(3, 2)$. Already in α.
- $(3, 2)$ and $(2, 4)$: gives $\mathbf{(3, 4)}$. **New!**

New pairs: $(1, 4)$ and $(3, 4)$.

Step 2: $\alpha^3 = \alpha^2 \circ \alpha$**.** The new second coordinates are all 4, but 4 has no outgoing edges in α, so no further pairs arise.

$$t(\alpha) = \alpha \cup \alpha^2 = \{(1, 2),\ (1, 4),\ (2, 4),\ (3, 1),\ (3, 2),\ (3, 4)\}.$$

Interpretation: $(x, z) \in t(\alpha)$ iff there is a directed path from x to z in the original digraph.



$$M_{t(\alpha)} = \begin{pmatrix} 0 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Bold = added 1's.

Reachability summary:

- From 1: can reach 2 and 4 (path: $1 \to 2 \to 4$).
- From 2: can reach 4.
- From 3: can reach 1, 2, and 4 (path: $3 \to 1 \to 2 \to 4$).
- From 4: no outgoing edges — 4 is a *sink*.